Linea 24 Technical and Service Guide

About this guide

This guide is for homeowners, installers and service technicians. It describes how the Linea 24 condensing combi boiler works, how to use its controls, what the error codes mean and how the boiler is serviced. Sections marked "for technicians" describe work that must be done only by a qualified gas engineer.

Read the warranty manual for the terms of the warranty, and the installer FAQ for installation details such as flue lengths and gas conversion.

Technical data

- Heating output: adjustable from 5.4 to 24.0 kW.
- Domestic hot water output: 24.0 kW.
- Hot water flow rate: 13.8 litres per minute with a temperature rise of 25 K.
- Seasonal space heating efficiency: 93%, energy class A. Hot water energy class A, load profile XL.
- NOx class: 6.
- Maximum heating circuit pressure: 3 bar. Minimum operating pressure: 0.5 bar.
- Expansion vessel: 8 litres, pre-charged to 1.0 bar.
- Heating flow temperature: 25 to 80 °C. Hot water temperature: 35 to 60 °C.
- Electrical supply: 230 V, 50 Hz. Power consumption: 82 W at full output, 3 W in standby.
- Dimensions: 700 mm high, 400 mm wide, 290 mm deep. Weight: 31 kg empty.
- Gas types: natural gas G20 as supplied, LPG G31 with conversion kit NW-KIT-31.

How the boiler works

The Linea 24 is a combination boiler: it heats the radiators and produces hot water on demand, without a hot water cylinder. A modulating fan mixes gas and air before the burner, so the flame adapts to the heat demand between about 20% and 100% of the maximum output.

The flue gases pass through the primary heat exchanger twice. The second pass cools them below their dew point, so the water vapour in them condenses and gives back its heat. This is why the boiler produces condensate, slightly acidic water that drains through the condensate trap and the condensate pipe. On a cold day it is normal to see a white plume of water vapour at the flue terminal.

When a hot water tap is opened, a flow sensor detects the flow and a diverter valve sends the heat to the plate heat exchanger for hot water. Hot water always has priority over heating. Heating resumes as soon as the tap is closed.

Controls and display

The control panel has a display, a heating knob, a hot water knob and a reset button.

- The display shows the flow temperature during heating, the hot water temperature while a tap is open, and the system pressure in bar when the boiler is idle.
- A flame symbol shows that the burner is lit. A radiator symbol or a tap symbol shows whether the boiler is heating the radiators or the hot water.
- An error code, such as E10, flashes when the boiler has stopped because of a fault.

Turn the heating knob fully anticlockwise to switch off heating for the summer. Hot water and frost protection keep working.

Heating and hot water settings

The heating knob sets the maximum flow temperature between 25 and 80 °C. For radiators a setting of 60 to 70 °C is usually enough. For underfloor heating, the installer limits the flow temperature to 45 °C with parameter P05.

The hot water knob sets the temperature between 35 and 60 °C. A setting of 48 °C gives comfortable hot water and reduces limescale in the plate heat exchanger.

Eco mode switches off the preheating of the plate heat exchanger. The boiler uses less gas, but hot water takes a few seconds longer to arrive at the tap. Press the hot water knob for 3 seconds to switch eco mode on or off; the display shows "ECO" while it is active.

If an NW-OS1 outdoor sensor is connected, the boiler adjusts the flow temperature to the outdoor temperature. The heating knob then shifts the heating curve up or down instead of setting a fixed temperature.

Frost protection

The boiler protects itself against frost as long as it has electricity and gas, even when heating is switched off. When the flow temperature falls below 7 °C the pump starts. If the temperature falls below 5 °C the burner lights and runs until the water reaches 15 °C.

Frost protection only protects the boiler itself, not the radiators, the pipes in unheated rooms or an external condensate pipe. If the property is left empty in winter, leave the heating on at a low room temperature or have the system drained.

System pressure

The heating system works under pressure. With the system cold, the display should show between 1.2 and 1.5 bar. The pressure rises by up to 0.5 bar when the system is hot; this is normal.

If the pressure is below 1.0 bar with the system cold, top it up:

1. Switch off the boiler and let the system cool down.
2. Open both valves of the filling loop under the boiler slowly until the display shows 1.5 bar.
3. Close both valves firmly and switch the boiler back on.

If you have to top up the pressure more than twice a year, there is probably a leak or a fault in the expansion vessel. Ask a service technician to check the system.

Error codes

When the boiler stops because of a fault, the display shows an error code. Codes marked "reset" can be cleared by pressing the reset button once. If the same code returns, call an Authorised Service Centre.

- E01, ignition failure: no flame detected after 3 ignition attempts. Check that the gas tap is open and that other gas appliances work, then reset.
- E02, overheating: the safety thermostat has tripped at 105 °C. Reset once the boiler has cooled down. Often caused by air in the system or a closed heating valve.
- E03, flue or air supply fault: the fan cannot reach its set speed, usually because the flue or the air intake is blocked. Do not reset repeatedly; call a technician.
- E10, low water pressure: the system pressure is below 0.5 bar. Top up the pressure to between 1.2 and 1.5 bar using the filling loop. The boiler restarts by itself.
- E20, fan fault: the fan does not turn or does not report its speed. Call a technician.
- E28, condensate trap blocked: the condensate cannot drain. In cold weather the condensate pipe is often frozen; pour warm, not boiling, water over the external part of the pipe, then reset.
- E35, flame detected with the gas valve closed. Call a technician; do not reset.
- E40, water temperature sensor fault. Call a technician.

Troubleshooting

No hot water, but heating works: check that the hot water knob is not at its minimum and that eco mode is not mistaken for a fault. If the hot water is warm but not hot, the plate heat exchanger may be scaled; this is common where the water is hard.

Radiators cold at the top: there is air in the radiators. Bleed them with a radiator key while the heating is off, then check the system pressure and top it up if needed.

The boiler is noisy or makes a kettle-like sound: this usually means limescale or sludge in the primary heat exchanger. Ask a service technician to clean the heat exchanger and to check the water treatment.

The pressure keeps dropping: look for drips at the radiator valves and under the boiler. Water dripping from the copper pipe of the safety valve outside the building means the pressure has gone above 3 bar, often because of a faulty expansion vessel.

Routine maintenance (for technicians)

The boiler must be serviced once a year. The extended warranty requires the annual service to be carried out by an Authorised Service Centre. Follow this procedure:

1. Measure the combustion with a flue gas analyser before opening the boiler. With natural gas, CO2 must be between 8.8 and 9.4% at maximum output and CO below 150 ppm.
2. Isolate gas and electricity and remove the front cover and the burner door.
3. Replace the burner door gasket and the insulation panel if they are damaged. The burner door gasket must be replaced every time the burner door is opened.
4. Clean the primary heat exchanger with the Northwind cleaning brush and rinse it with

water. Never use a wire brush or acid.

5. Inspect the ignition and detection electrodes. Replace them if the gap is outside 3.5 to 4.5 mm or if they are worn. In normal use they last about 3 years.
6. Remove, clean and refill the condensate trap with water before reassembly.
7. Check the expansion vessel pre-charge with the system drained to zero pressure: 1.0 bar.
8. Reassemble, test for gas tightness, measure the combustion again and fill in the maintenance booklet.

Spare parts and service life

Use only original Northwind spare parts. The parts most often replaced during the life of the boiler are the ignition electrode, the condensate trap seal, the diverter valve cartridge and the pump. The expected service life of the boiler is 15 years with annual maintenance.

Safety

If you smell gas, do not switch lights or electrical appliances on or off. Close the gas tap at the meter, open the windows, leave the building and call the gas emergency number from outside.

Never cover the boiler, block the flue terminal or obstruct the air intake. Do not remove the casing: there are no user-serviceable parts inside.

Disposal

At the end of its life the boiler must be taken to an authorised collection centre for electrical and electronic equipment. Do not dispose of it with household waste. The installer who replaces the boiler can usually take the old one away.